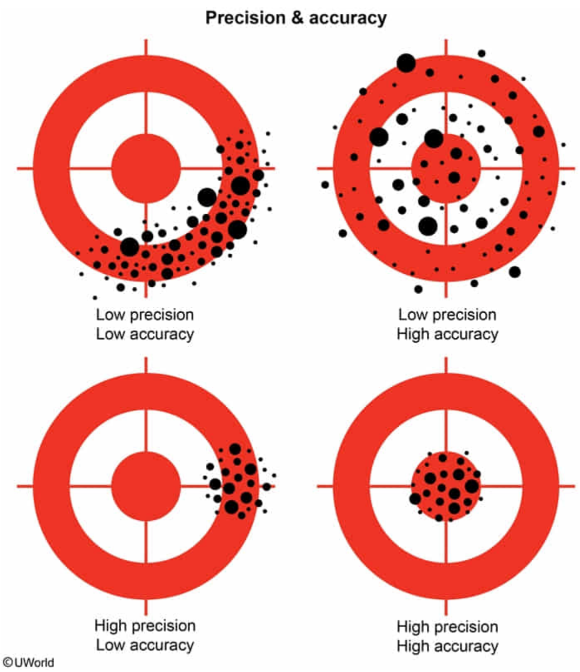
The affinity of a protein for its ligand was measured by isothermal titration calorimetry. The experiment was repeated multiple times to mitigate error by:

- A. assessing the precision of the results. [50%]
- B. verifying the calibration of the instruments. [4%]
- C. assessing the validity of the experiment. [13%]
- D. verifying the accuracy of the results. [31%]

Correct
50% answered correctly

Explanation:



A measurement is said to be **precise (reliable)** if similar results are obtained on repeated trials. It is **accurate** if it correctly represents the **real-world value** as assessed by comparison with a "**gold standard**," the best available benchmark test that is considered accurate. One or more gold standards are used to **calibrate** the instrument being used for measurement, allowing researchers to be confident that future measurements are accurate. If experimental results are both **precise and accurate**, they are said to be **valid**, as they reliably represent the correct value.

In this scenario, an experiment was repeated multiple times. This allowed researchers to determine whether repeated trials yield similar results. Therefore, the purpose of the repetition was to **assess the precision** of the results.

(Choices B and D) To assess accuracy, the instrument used for measurements must be calibrated against a gold standard. Because no gold standard calibration is mentioned, accuracy of the results cannot be determined.

(Choice C) Valid experiments must have results that are both precise *and* accurate because they must correctly measure the real-world value of the phenomenon of interest. Because the results were not compared to a gold standard, accuracy cannot be determined.

Educational objective:
A measurement is said to be precise, or reliable, if similar or very close results are obtained on repeated trials. Accuracy can be assessed using a "gold standard" test, which is used to calibrate instruments. A result is said to be valid if it is both precise and accurate, as it reliably produces real-world values.

Time spent: 0 sec	**QID**:402109
Subject: Biochemistry	**System:** Amino Acids and Proteins